Lecture 02

Water Resources Systems Modeling

Review on Lecture 01

- ❖ Overview of WRSA
  - ◆ A water resources system is the whole made from connected ***hydrologic, infrastructure, ecologic, and human processes that involve water***. (Brown et al. 2015)
- ❖ WRSA- ***"Wicked"*** Problems?
  - ◆ "**Defining** the problem is the problem itself"
- ❖ 23 Unsolved Problems in Hydrology (UPH)
  - ◆ "Why" "What" & "How" questions
  - ◆ Real-world Example of WRSE problem: 보령 다목적댐
- ❖ **Future of WRSA? (Reed & Kasprzyk 2009)**
  - ◆ **Constructive Modeling … *Interfaces with society***

Before Lecture 02

❖ **<u>Acronyms (약어)/Keywords</u>** in Lecture 02

◆ **DSS**: Decision Support Systems (의사결정 지원 시스템)
= computerized program used to support determinations, judgments, and courses of action in an organization or a business

◆ **SVP**:
Shared Vision Planning

◆ **SVPM**:
Shared Vision Planning Model

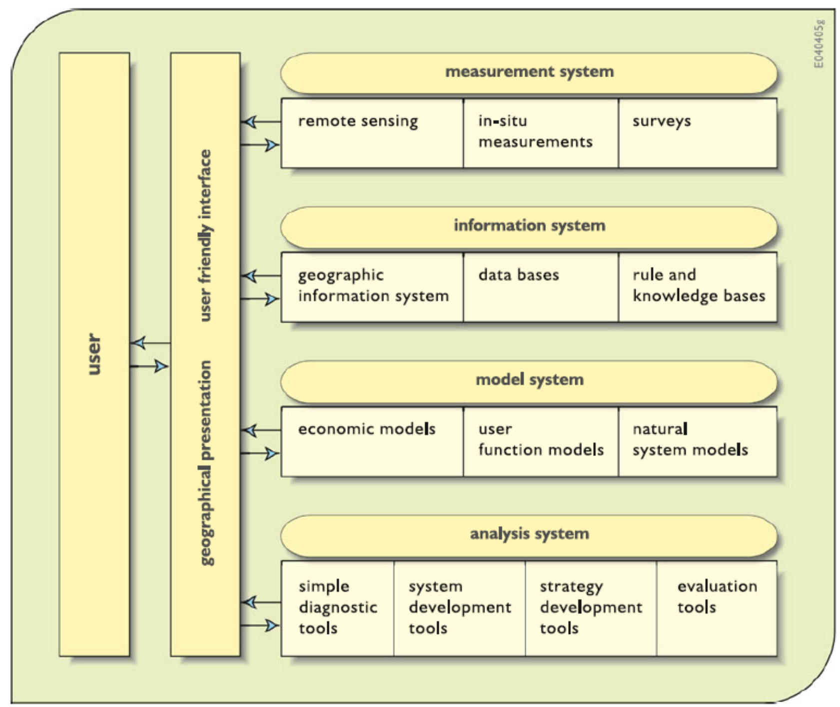


Fig. 2.6 Common components of many decision support systems

Today's Topic: WRS Modeling

- ❖ Modeling Water Resources Systems
- ❖ Types of Water Resources Systems Models
- ❖ Examples of Water Resources Systems Models

- ❖ Training with Systems Modeling (LAPTOP)
  - ◆ Assignment #1의 #4-C에서부터 실제 보령댐 저수지 운영 모델을 Microsoft Excel을 통해서 만들어 볼 예정

Modeling Water Resources Systems

❖ Models
= **<u>"Simplified Representations"</u>** of the real-world systems
(실제 시스템의 단순화된 표현방법, 모델링을 하는 목적)

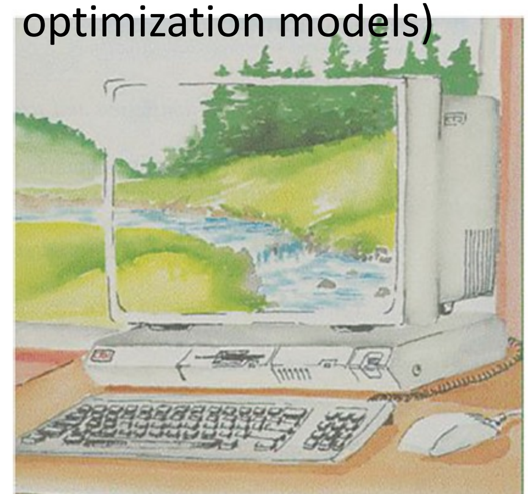
◆ 모델을 사용하는 이유 (목적)
예측: "내가 지금 내리는 결정이 실제로 어떤 결과를 가져올까?"

Qualitative information & beliefs

Fig. 2.1 Using mental models for prediction

Quantitative mathematical models (Mathematical simulation or optimization models)

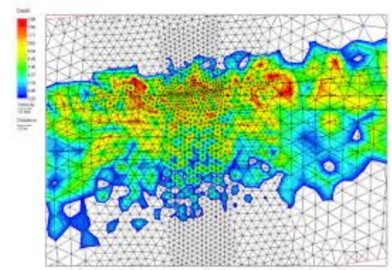
Fig. 2.2 Using computer-based mathematical models for prediction

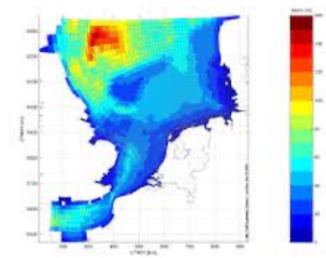


Modeling Water Resources Systems

❖ Limitations with Modeling Approach

- ◆ Complexity <복잡성> … Why?
  - Computational Limitations? NOT SO MUCH TRUE IN WRS MODELING (Although may hold true in some problems)
  - Hard to understand the **<u>multiple interdependent physical, biochemical, ecological, social, legal, and political (human) processes</u>** that govern the behavior of such water resource systems





- ◆ Requires numerous <u>approximations</u> and <u>assumptions</u>

=> 해결하고자 하는 문제의 범위나 문제를 해결/모델링하기 위한 가정들을 명확하게 설정하는 것이 매우 중요함!
(related to space variability & scaling in 23 UPHs)

Modeling Water Resources Systems

❖ Challenges in Modeling

◆ Challenges of <u>Planners and Managers</u> (의사결정자)

- 실제로 물 문제를 해결하는 사람 (정책을 만들고 정책을 시행)
- Ex) 수자원공사, 한국수력원자력, 등등의 기관 소속

◆ Challenges of <u>Modelers</u>

- 모델을 만들고 모델을 사용하여 수치적인 분석을 진행하는 사람
- 의사결정자나 실제 사용자들의 의견을 받아 이를 모델로 구현하는 역할
- 주로, 학계나 연구직에 종사하는 전문가가 많음

◆ Challenges of <u>Applying Models in Practice</u>

- 실제로 물 문제를 경험한 사람
- 의사결정자나 모델을 개발한 사람들보다 상대적으로 전문 지식이 부족할 수 있음

What constitutes WRS Models?

<u>* Components consisting a Model ***</u>**

◆ Parameters (매개변수):
variables that are assigned <u>KNOWN</u> values

◆ Decision Variables (결정변수): variables having unknown values that are to be determined by solving the model

- Design variables: 저수지의 용량, 보의 높이 등 설계와 관련된 변수들
- Operating variables: 저수지 방류량, 용수 배분량 등

◆ State Variables (상태변수):
variables that describe the state of the system

◆ Constraints (제약조건): the conditions the system has to satisfy

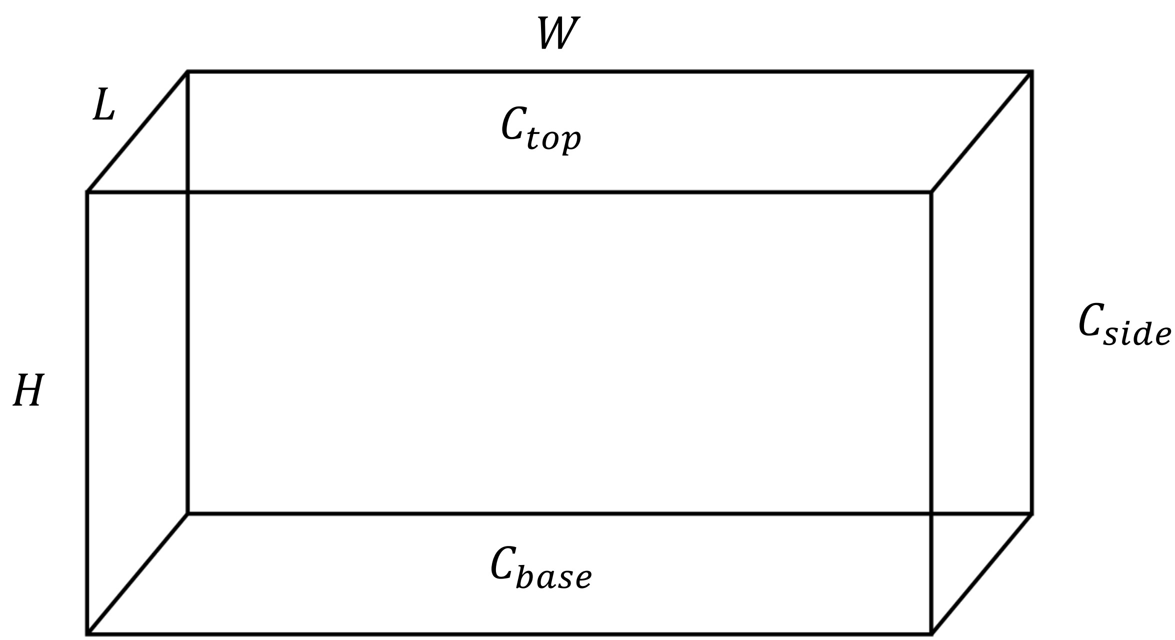
◆ (+) Objective Function (목적함수) (in case of optimization models):
function to be optimized during the problem solving procedures

Example of the Model Formulation

$$Minimize\ Cost$$

$$Cost = \left(C_{base} + C_{top}\right)(LW) + 2(C_{side})(LH + WH)$$

Subject to: $LWH \geq V$ Determine W, L, and H.



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❖ **Objective Function** (목적함수)

- Cost function, Loss function, Utility function 등으로 불릴 수 있음
- 계산 과정을 통해 반복적으로 계산되는 함수를 뜻함

❖ **Parameters** (매개변수) = variables that are assigned known values

❖ **Decision Variables** (결정변수)
= variables that are to be determined by solving the model

❖ **Constraints** (제약조건) = the conditions that the systems has to satisfy

Types of Water Resources Systems Models

❖ Optimization Models

◆ REQUIRES an objective function to be maximized/minimized

◆ Solution = Values of ALL of Unknown Decision variables that satisfy the predefined constraints

❖ Simulation Models: Answers "What if" questions

◆ What will likely to happen under a particular design or operating policy?

◆ Works well when there are only a relatively few alternatives to be evaluated (trial-and-error process)

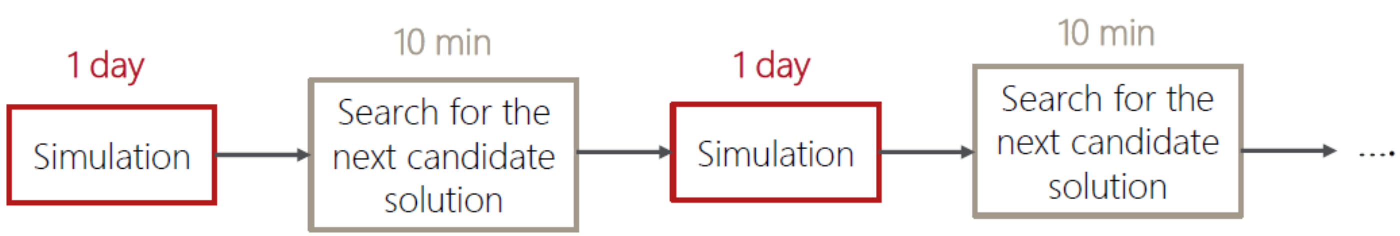
❖ How about **Simulation-Optimization Models?**

Types of Water Resources Systems Models

❖ **Simulation Optimization Models (Simulation/Optimization) (=Simulation-based optimization)**

- ◆ Integrates optimization techniques into simulation modeling & analysis (usually with stochastic objective functions)
- ◆ Examples: Heuristic methods (genetic algorithms) & Dynamic programming models

- ◆ Primary Limitation = Computational Expenses
  Simulation optimization is computationally expensive
  (Solution: Parallel Computing)

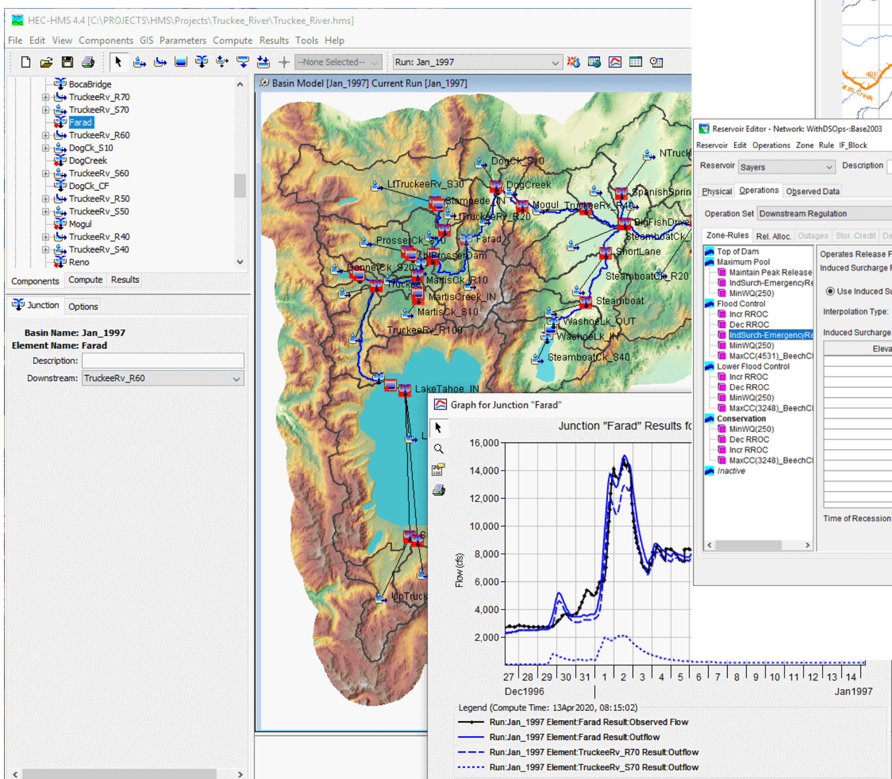


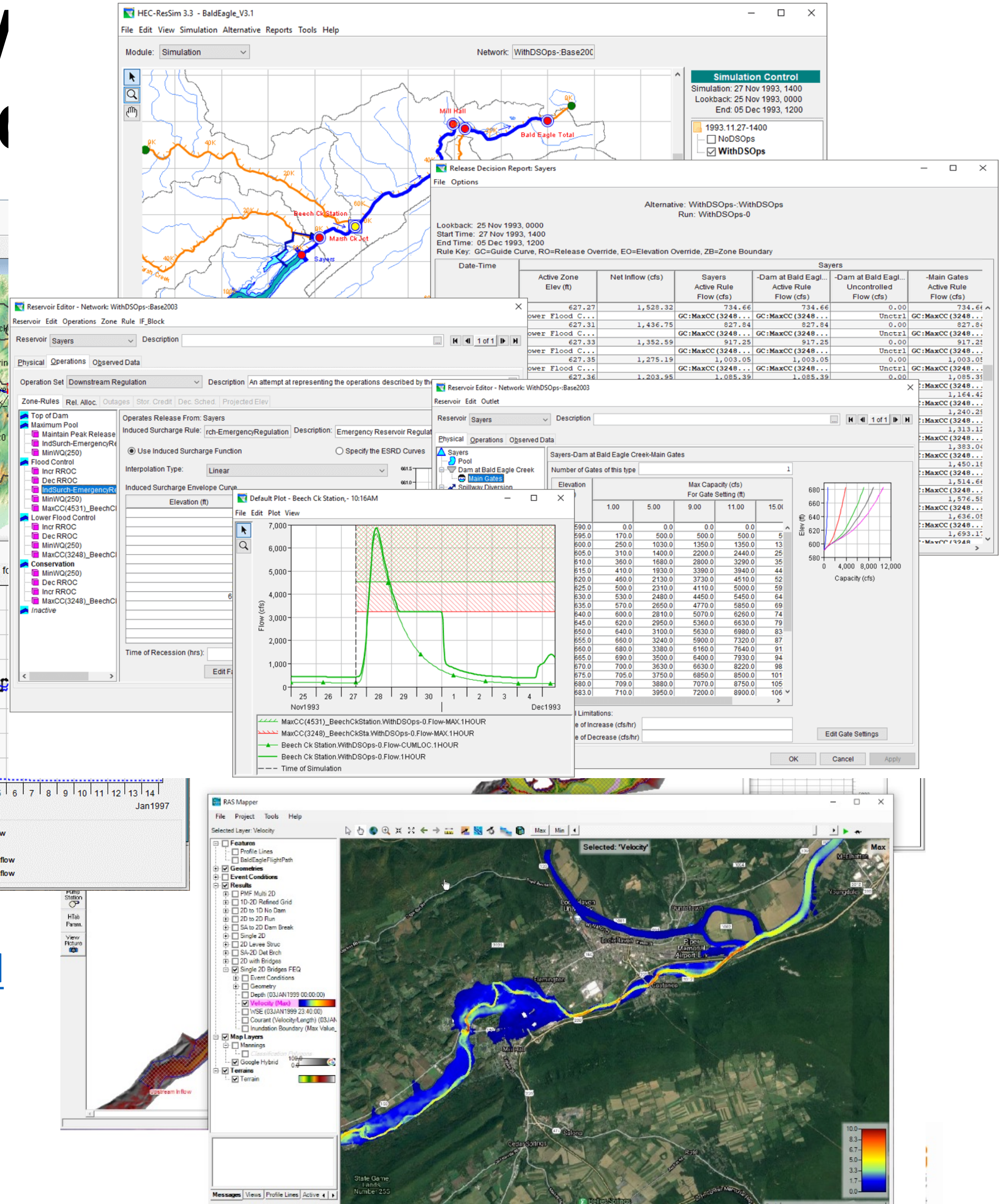
Examples of Water Resources Systems Models Applied in Practice (Industry)

❖ US Army Corps of Engineers (미군 육군 공병대)- Hydrologic Engineering Center (HEC)

- **HEC-HMS (Hydrologic Modeling System)**
  - Traditional hydrologic analysis procedures (event infiltration, unit hydrographs, hydrologic routing)
  - Continuous simulation (evapotranspiration, snowmelt, soil moisture accounting, runoff simulations)
- **HEC-RAS (River Analysis System)**
  - 1-D steady flow, 1&2-D unsteady flow calculations
  - Sediment transport/mobile bed computations
  - Water temperature & water quality modeling
- **HEC-ResSim (Reservoir Simulation)**
  - Model reservoir operations for a variety of operational goals and constraints

Examples of W
Models Applie





https://www.hec.usace.army.mil/software/default.aspx

Modeling Water Resources Systems

- ❖ Challenges in Modeling
  - ◆ Challenges of Planners and Managers (의사결정자)
    - 실제로 물 문제를 해결하는 사람 (정책을 만들고 정책을 시행)
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  - ◆ Challenges of Modelers
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    - 실제로 물 문제를 경험한 사람
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Fig. 2.5 Stakeholders involved in river basin planning and management, each having different goals and information needs

Modeling in the Real-world: Example

❖ Application of **<u>Shared Vision Planning</u>** in Korea

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Elicitation of drought alternatives based on Water Policy Council and the role of Shared Vision Model

Kim, Gi Joo[a]* · Seo, Seung Beom[b] · Kim, Young-Oh[c]

[a]Ph.D student, Department of Civil & Environmental Engineering, Seoul National University, Seoul, Korea
[b]Research Fellow, Division for Integrated Water Management, Korea Environment Institute, Sejong, Korea
[c]Professor, Department of Civil & Environmental Engineering, Seoul National University, Seoul, Korea

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협의체 기반 가뭄 대응 대안 도출과 비전공유모형의 역할

김기주[a]* · 서승범[b] · 김영오[c]

[a]서울대학교 공과대학 건설환경공학부 박사과정, [b]한국환경정책·평가연구원 통합물관리연구실 부연구위원, [c]서울대학교 공과대학 건설환경공학부 교수

요 지

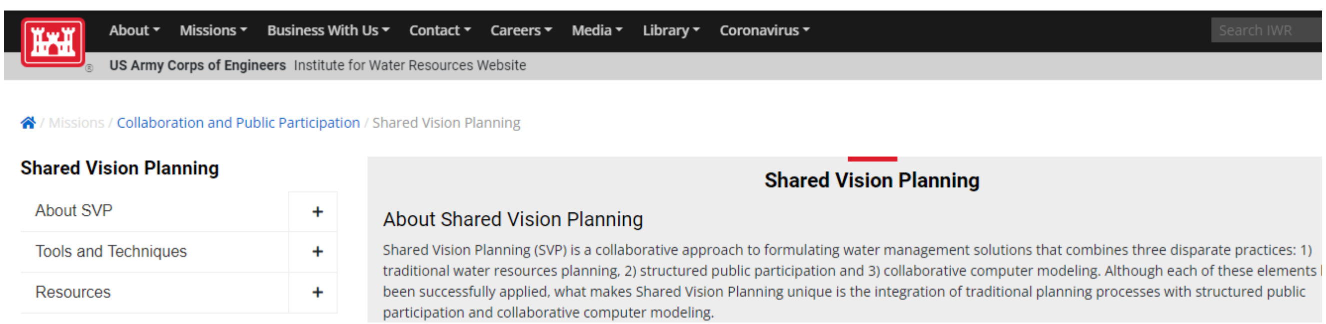
기후변화로 인한 다년 가뭄은 전세계적으로 증가하는 추세이며, 충청남도에 위치한 보령댐 또한 2014년부터 2017년까지 지속된 다년 가뭄으로 인해 큰 피해를 받았다. 다양한 가뭄 피해 저감 정책 설립 과정에 있어 일방적인 하향식 의사결정 과정을 바탕으로 진행된 정책은 이해당사자간의 갈등을 야기했기에, 이를 방지하기 위해서는 이해당사자와 정책결정자들간의 참여형 의사결정 과정이 필수적이다. 본 연구에서는 다양한 그룹으로 구성된 이해당사자의 참여를 독려하는 참여형 의사결정 방식 중 하나인 비전공유계획을 충청남도 기후변화 적응 물관리정책 협의회를 통해 체계적으로 적용하였다. 또한, 비전공유계획의 핵심 요소인 비전공유모형을 시스템 다이내믹스 모형으로 개발하였고, 총 3회의 소위원회를 거쳐 이해당사자의 요청사항에 맞추어 모형을 보완하였다. 구축한 모형을 활용하여 미래에 발생 가능한 가뭄의 위험을 포함하고 있는 기후변화 시나리오로 모의하였고, 보령댐과 보령댐 계통 지자체의 가뭄으로 인한 취약성을 빈도, 지속기간, 크기 개념의 평가지표로 표현하였다. 모의 결과, 용수 공급원인 보령댐은 용수 수급처인 지자체보다 가뭄에 상대적으로 더 취약하며, 8개 지자체 중 가뭄 대응 대책이 주로 계획되어 있는 지역과 모의에서 추정한 가뭄 발생 지역이 일치하지 않음을 확인하였다. 모의 결과를 향후 협의회 회의에서 이해당사자와 공유하고, 댐과 지자체의 입장에서 용수 부족을 해소할 방안을 모형에 적용함으로써 미래 정책 결정 및 갈등 해소를 위해 개발한 비전공유모형을 이용할 수 있음을 제안하였다.

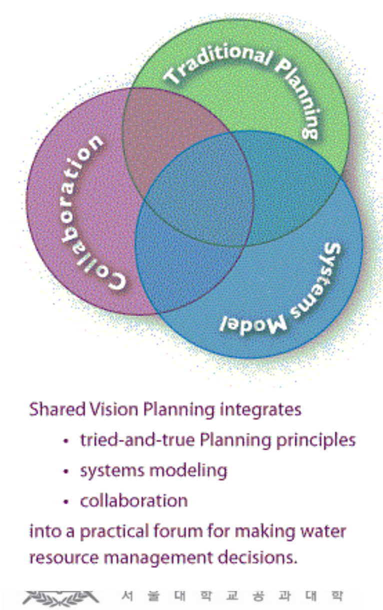
Modeling in the Real-world: Example

❖ Shared Vision Planning (비전공유계획) (교재 p.65-67)

- ◆ Type of an **<u>Interactive Model-building Technique</u>**
  - Stakeholders hardly believe in the model already built (and this is partially TRUE- you will see in the exercise)
- ◆ **Objective**:

  Involving Stakeholders in Model Building Procedures in order to give them a feeling of ownership





Modeling in the Real-world: Example

❖ Shared Vision Planning (비전공유계획)

History of Shared Vision Planning in the Army Corps of Engineers

In 1989, on the heels of severe droughts in much of the West, Southeast and the Missouri-Mississippi Valley, the Corps of Engineers began the National Drought Study to find a better way to manage water for drought. After a year of study and collaboration with many groups focused on drought that year, the Corps proposed a drought preparedness method and applied it in test cases around the country. The method was a form of the systems analysis approach designed during the Harvard Water Program of the late 1950s and early 1960s that later became the basis for Federal water resources planning, "Principles and Standards" (1973) and "Principles and Guidelines" (P&G, 1983). The Drought Preparedness Method, however, went further than the P&G. It required planners to find out what criteria decision makers and stakeholders would use in accepting or rejecting a drought plan and then develop metrics so that each alternative could be evaluated according to those criteria.

In 1991, Richard Palmer, now a civil engineering professor at the University of Washington, attended a workshop of the Cedar and Green River Case Study in Seattle (a case study of the ongoing National Drought Study). There he proposed that the Corps develop system simulation models in each test case and showed how the models could be built with stakeholders and decision makers. Each of the five case study managers (for the five National Drought Study case studies) agreed to do so, although they were not allowed any increase in budget or time for what might appear to be an “extra” task. At the time, Dr. Palmer was using an object-oriented software called STELLA®, which made it easier to create models that could be understood by non-modelers because the functional relationships were diagrammed, as they were mathematically defined. Stakeholders could literally see the factors that affected any variable. Reservoir systems appeared as a series of boxes connected by flows.

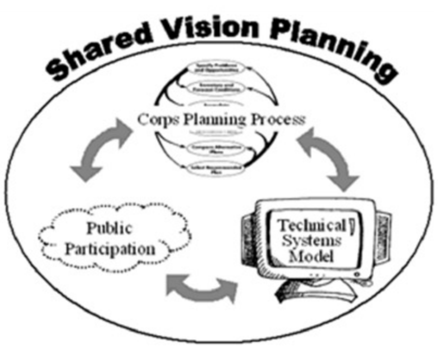


Fig. 1. Three pillars of Shared Vision Planning (USACE, 2010)

https://www.iwr.usace.army.mil/Missions/Collaboration-and-Conflict-Resolution/SVP/About-SVP/History-of-Shared-Vision-Planning/

Modeling in the Real-world: Example

❖ 충청남도 기후변화 적응 물관리정책 협의회

- ◆ 불확실성이 큰 **기후변화 적응의 기본 철학 아래**에서
- ◆ 다양한 이해관계를 가진 사람들이 투명한 정보를 공유하고 신뢰를 바탕으로 서로 배우며
- ◆ 비전을 만드는 시작 단계부터 대안을 선정하고 실현하여 평가하는 마지막 단계까지 전 과정을 함께 하여 합의를 이끌어내기 위함.

❖ Timeline

- ◆ 2016-12-07 창립총회 개최
- ◆ 2017-09-21 1차 소위원회
- ◆ 2017-11-03 2차 소위원회
- ◆ 2018-07-25 3차 소위원회

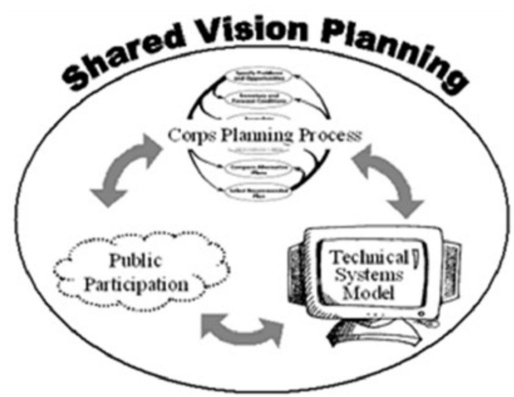
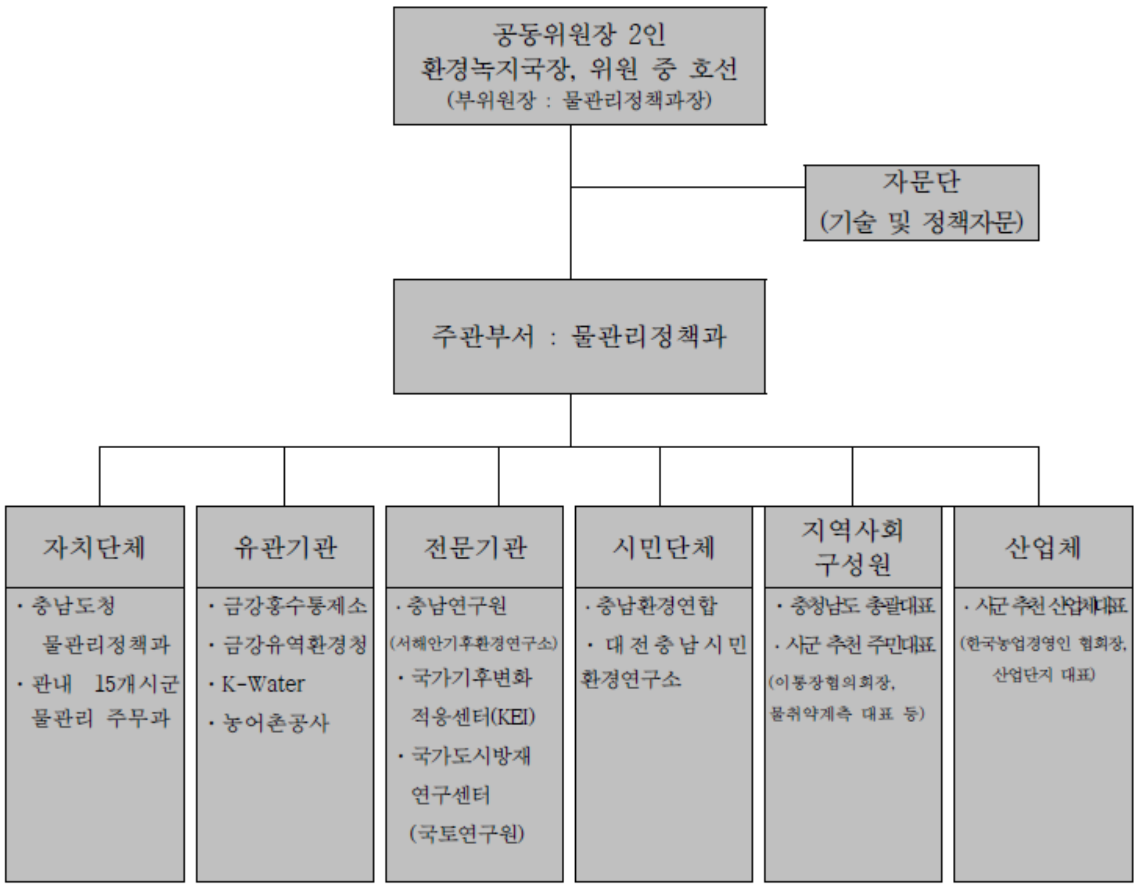


Fig. 1. Three pillars of Shared Vision Planning (USACE, 2010)

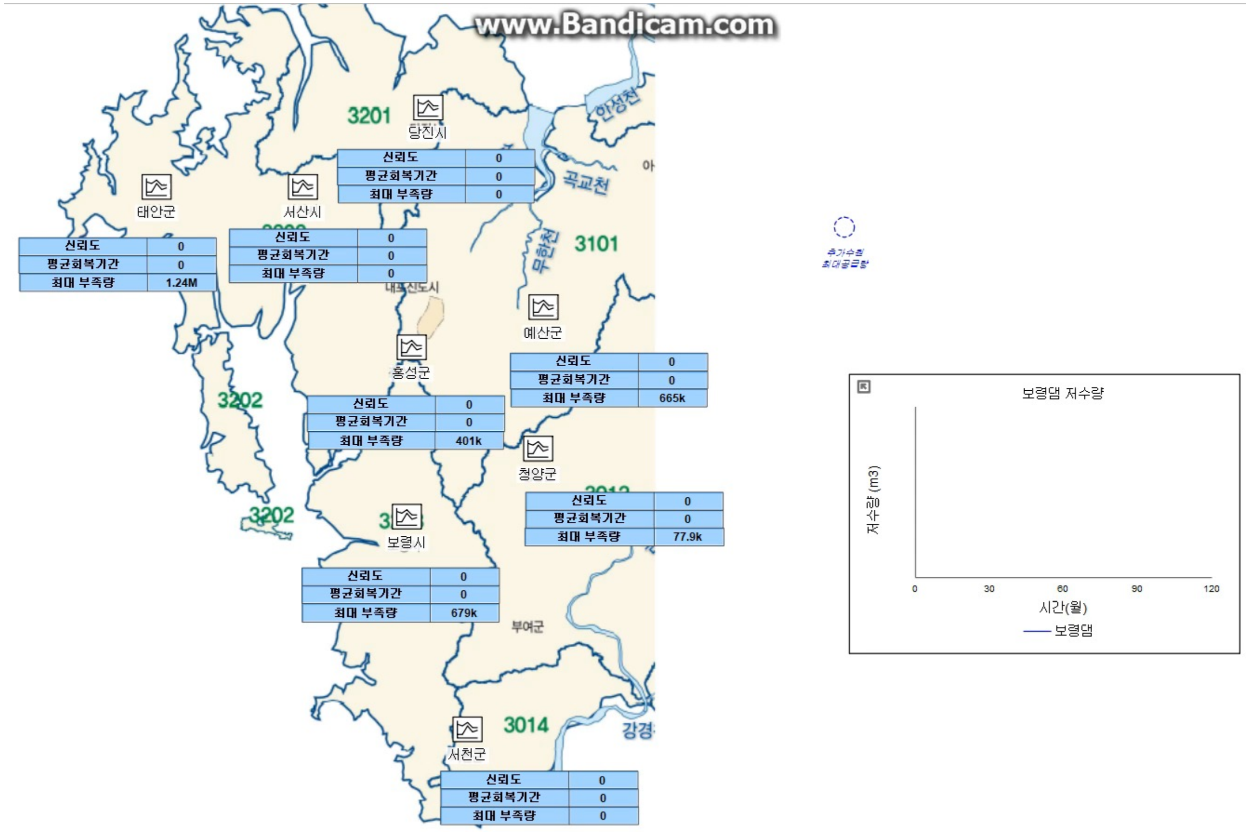
Modeling in the Real-world: Example

충청남도 기후변화 적응 물관리정책(수자원분야) 협의회 구성표(제4조제1항 관련)



Modeling in the Real-world: Example

❖ 1차 소위원회 (2017-09-21) 사용 모형



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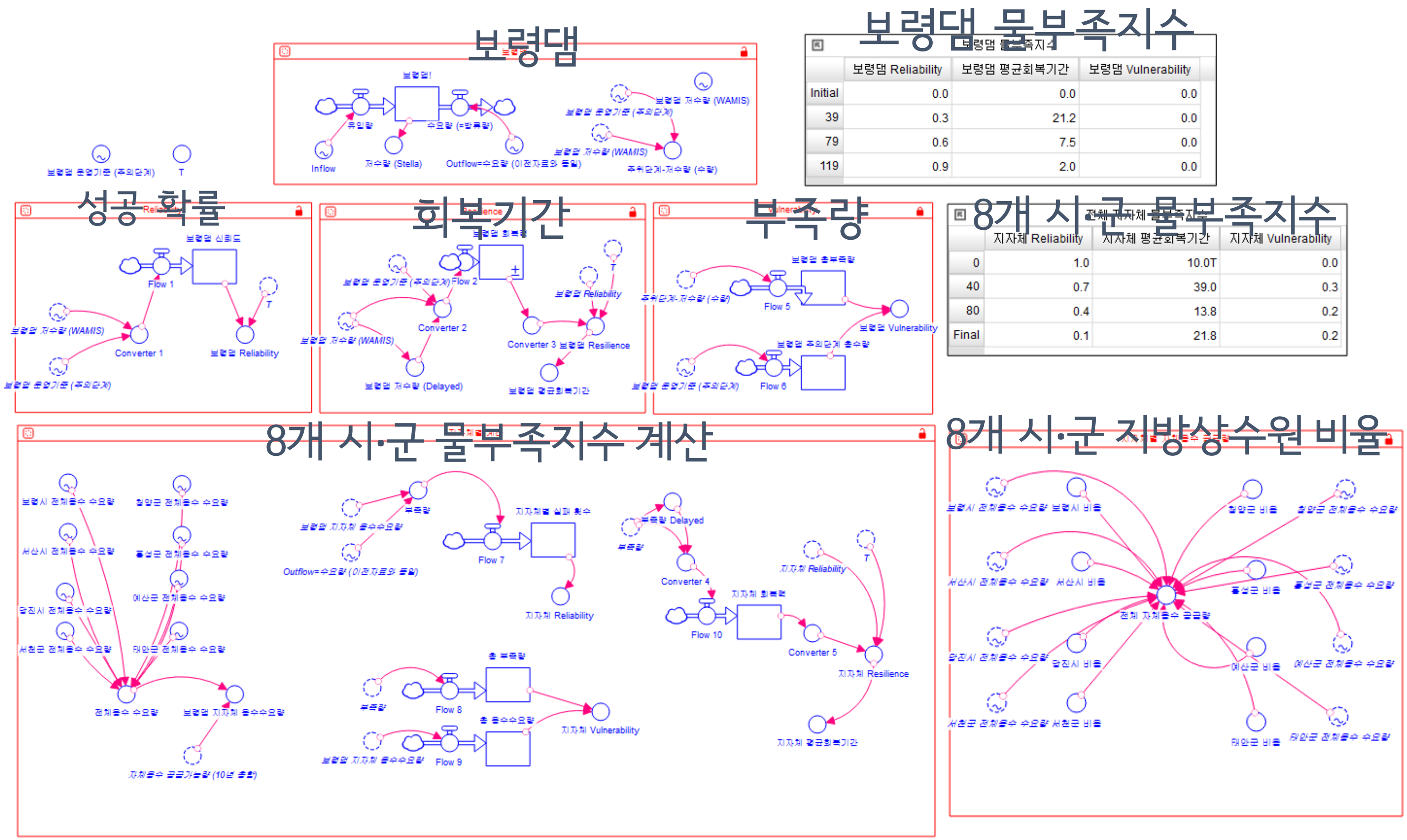
Modeling in the Real-world: Example

❖ 2차 소위원회 (2017-11-03) 사용 모형

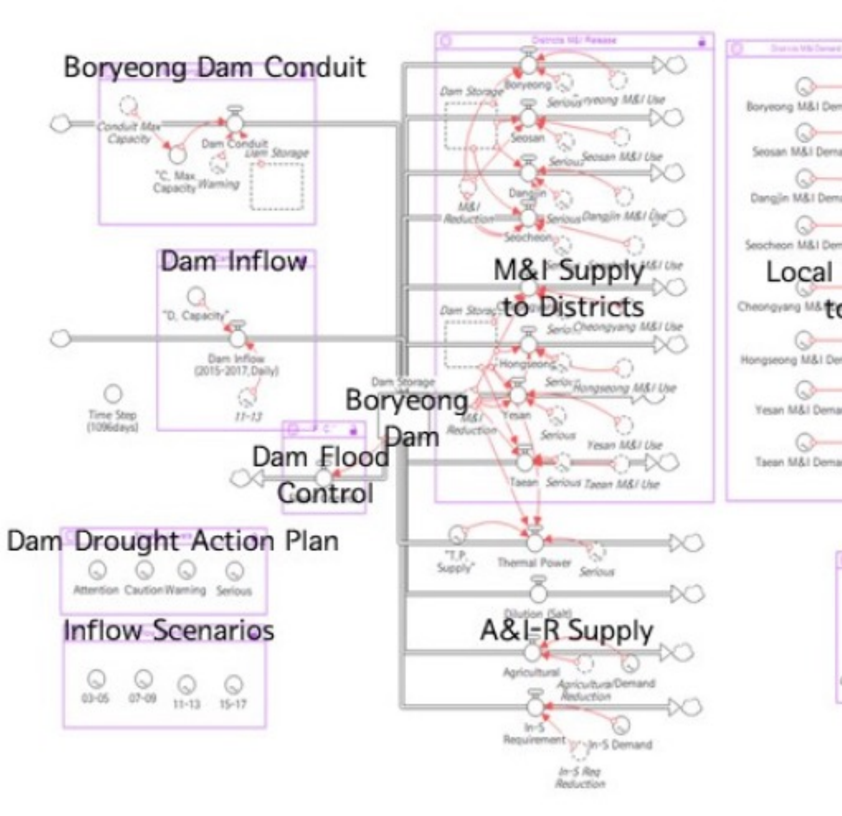


	보령댐 Reliability	보령댐 평균회복기간	보령댐 Vulnerability
Initial	0.0	0.0	0.0
39	0.3	21.2	0.0
79	0.6	7.5	0.0
119	0.9	2.0	0.0

	지자체 Reliability	지자체 평균회복기간	지자체 Vulnerability
0	1.0	10.0T	0.0
40	0.7	39.0	0.3
80	0.4	13.8	0.2
Final	0.1	21.8	0.2

Modeling in the Real-world: Example

❖ 3차 소위원회 (2018-07-25) 사용 모형



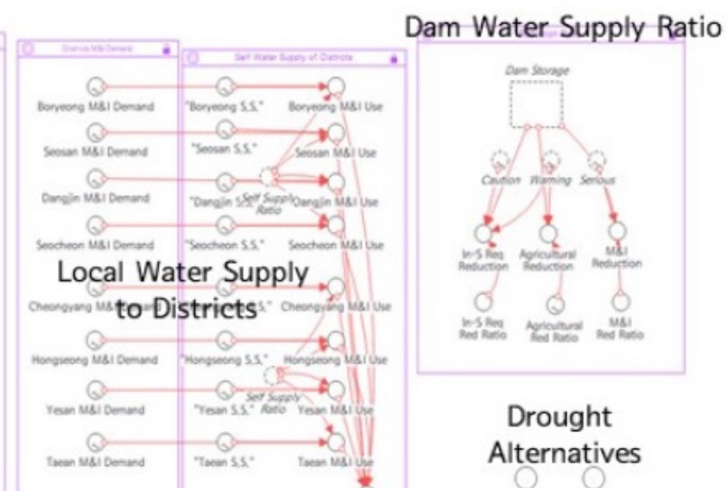


Fig. 4. The STELLA Shared Vision Model for Bo

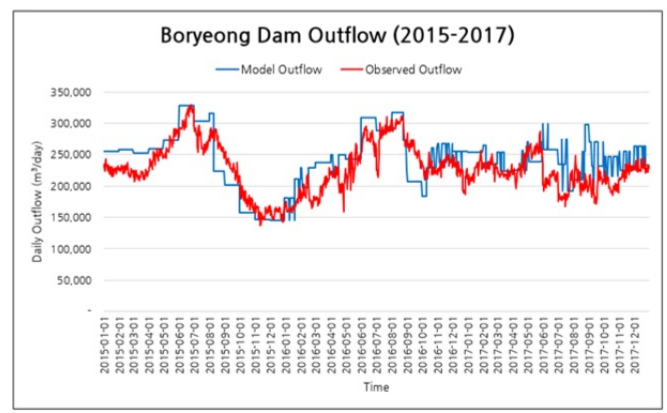


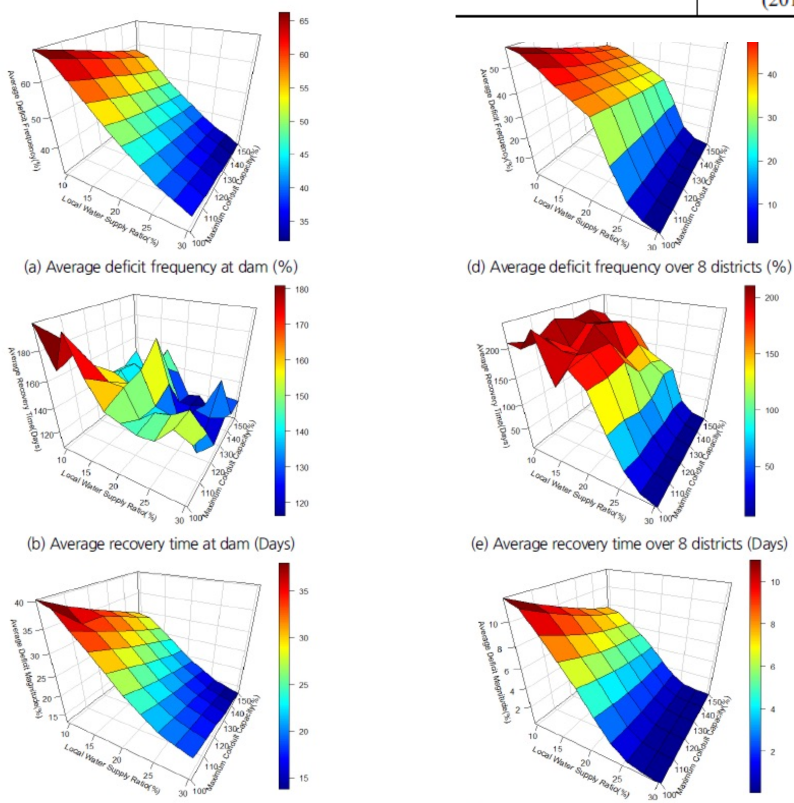
Fig. 5. Boryeong Dam outflow comparison: simulated vs. observed

Modeling in the Real-world: Example

❖ 모형을 사용한 분스

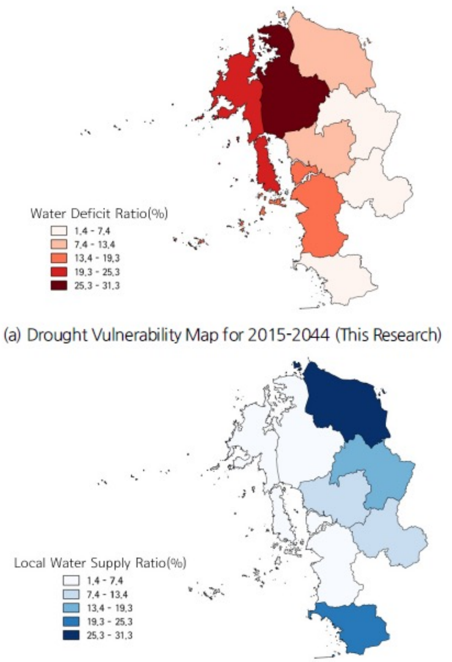
Table 4. Inflow and demand scenarios of each run

Inflow/ Demand Scenarios \ Run	#1	#2	#3	#4
Inflow Scenario	Average Inflow (Observed, 2003-2013)	Drought Inflow (Observed, 2015-2017)	Drought Inflow (Observed, 2015-2017)	GCM Scenarios (Seo and Kim, 2018)
Demand Scenario	Actual Water Demand (2015-2017)	Actual Water Demand (2015-2017)	Projected Water Demand (2015-2044)	Projected Water Demand (2015-2044)



(a) Average deficit frequency at dam (%)
(d) Average deficit frequency over 8 districts (%)
(b) Average recovery time at dam (Days)
(e) Average recovery time over 8 districts (Days)
(c) Average deficit magnitude at dam (%)
(f) Average deficit magnitude over 8 districts (%)

Fig. 6. Response function for the worst GCM scenario



(a) Drought Vulnerability Map for 2015-2044 (This Research)

(b) Local Water Supply Ratio by Districts for 2018-2025 (Chungcheongnam-do, 2018)

Fig. 7. Comparison of future drought vulnerability and future alternatives of districts

Now, It's Your Turn to build a model.

❖ Build a reservoir simulation model assuming known inflow series I_t. What are the governing equations and the constraints? Assuming that you are the dam operator, what might be a possible objective function? How about when you are a water user? Which objective between the two is more important? (refer to the equations in slide # 10).

** Flow Diagram of a Reservoir Simulation Model **

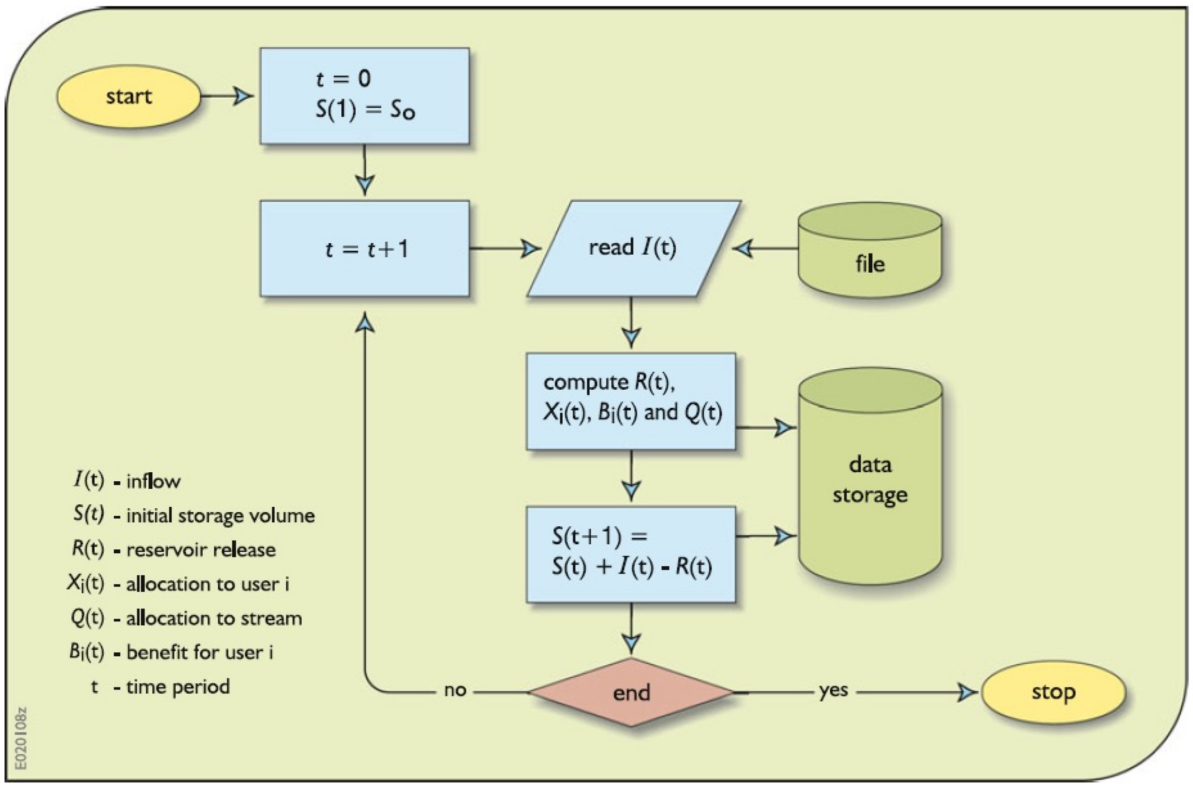


Fig. 3.5 Flow diagram of the reservoir—user allocation system simulation process. The simulation terminates after some predefined number of simulation time steps

Conclusion of Lecture 02

- **Modeling Water Resources Systems**
  - Models = <u>**"Simplified Representations"**</u> of the real-world systems
  - We use models to predict (or answer "what if" questions) possible consequences (especially <u>**quantitative**</u>) of our actions

- **Types of Water Resources Systems Models**
  - Optimization, Simulation, Simulation/Optimization Models

- **Examples of Water Resources Systems Models**
  - HEC-HMS, HEC-RAS, HEC-ResSim Models
  - Shared Vision Planning Model

Future Schedule

❖ 다음 주 강의 (2022-03-22 화요일 17:00-19:50)

◆ Lec 03: Introduction to Conventional Optimization Models (DP&LP) Part I

- **General Concepts of Optimization Models**
- **Concepts of Linear Programming Models (LP)**
- **Examples of Problem Solving with LP Models**
- Concepts of Dynamic Programming Models (DP)
- Examples of Problem Solving with DP Models
- Real-world Applications of Traditional Optimization Models

◆ Assignment #1 DUE Before Class on Mar-22 (출력본 제출)

◆ Quiz #1 on Mar-29 // Datacamp #1 Due Mar-29

Real-world Reservoir Simulation

❖ Assignment #1의 4-5번 문제를 엑셀로 수행.

❖ What are the key assumptions you made in building the model? List at least three key assumptions inside the model.

❖ How will you enhance the model's practicality? (어떻게 모델을 더욱 현실과 비슷하게 개선할 것인지?)